

LAB 5 JAVA

1. Write a Java program that demonstrates method overriding by creating a superclass called Animal and two subclasses called Dog and Cat. • The Animal class should have a method called make Sound), which simply prints "The animal makes a sound." • The Dog and Cat classes should override this method to print "The Cat/The dog meows/barks" respectively. • The program should allow the user to create and display objects of each class.

[Hint: Use multilevel inheritance]

```
AnimalMain.java ×
AnimalMain.java > Language Support for Java(TM) by Red Hat > Dog > sound()
1  class Animal {
2      void sound() {
3          System.out.println(x: "The animal makes a sound.");
4      }
5  }
6
7  class Cat extends Animal {
8      @Override
9      void sound() {
10         System.out.println(x: "The Cat meows");
11     }
12 }
13
14 class Dog extends Cat {
15     @Override
16     void sound() {
17         System.out.println(x: "The dog barks");
18     }
19 }
20
21 public class AnimalMain {
22     Run main | Debug main | Run | Debug
23     public static void main(String[] args) {
24
25         Animal a = new Animal();
26         a.sound();
27
28         Cat c = new Cat();
29         c.sound();
30
31         Dog d = new Dog();
32         d.sound();
33     }
34 }
```

Output:

```
> cd "c:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise\" ;
lMain.java } ; if ($?) { java AnimalMain }
The animal makes a sound.
The Cat meows
The dog barks
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise>
```

2. Create one superclass HillStations and three subclasses Manali, Mussoorie, Gulmarg. Subclasses extend the superclass and override its location () and famousFor() method.

i. call the location () and famousFor() method by the Parent class', i.e. Hillstations class. As it refers to the base class object and the base class method overrides the superclass method; the base class method is invoked at runtime.

ii. call the location () and famousFor() method by the all subclass',and print accordingly.

```
J Main.java 1 X
Hill > J Main.java > Java > Main > main(String[] args)
1  class HillStations {
2      void location() {
3          System.out.println(x: "Location is:");
4      }
5
6      void famousFor() {
7          System.out.println(x: "Famous for:");
8      }
9  }
10
11 class Manali extends HillStations {
12     @Override
13     void location() {
14         System.out.println(x: "Manali is in Himachal Pradesh");
15     }
16
17     @Override
18     void famousFor() {
19         System.out.println(x: "It is Famous for Hadimba Temple and adventure sports");
20     }
21 }
22
23 class Mussoorie extends HillStations {
24     @Override
25     void location() {
26         System.out.println(x: "Mussoorie is in Uttarakhand");
27     }
28
29     @Override
30     void famousFor() {
31         System.out.println(x: "It is Famous for education institutions");
32     }
33 }
34
35 class Gulmarg extends HillStations {
36     @Override
37     void location() {
38         System.out.println(x: "Gulmarg is in J&K");
39     }
40
41     @Override
42     void famousFor() {
43         System.out.println(x: "It is Famous for skiing");
44     }
45 }
46
47 public class Main {
48     Run main | Debug main | Run | Debug
49     public static void main(String[] args) {
50
51         // Calling by Parent class reference
52         HillStations h = new HillStations();
53         h.location();
54         h.famousFor();
55
56         // Calling by all subclasses using parent reference
57         h = new Manali();
58         h.location();
59         h.famousFor();
60
61         h = new Mussoorie();
62         h.location();
63         h.famousFor();
64
65         h = new Gulmarg();
66         h.location();
67         h.famousFor();
68     }
69 }
```

Output:

```
PROBLEMS 16 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Main.java } ; if ($?) { java Main }
Location is:
Famous for:
Manali is in Himachal Pradesh
It is Famous for Hadimba Temple and adventure sports
Mussoorie is in Uttarakhand
It is Famous for education institutions
Gulmarg is in J&K
It is Famous for skiing
PS C:\Users\raksh\Desktop\ADJ Audip\Java\src\Practise\Hill>
```